Wentao Xu

Cory Hall 406, Berkeley, CA 94720
wentao.xu@berkeley.edu

Education

University of California, Berkeley — Berkeley, U.S.A.
PhD in Electrical Engineering and Computer Sciences, GPA: 4.0/4.0 — Sep. 2023 - Present

- Advisor: Prof. Jessica Boles*
- MS, thesis title: "Overtone Piezoelectric Resonators for Power Conversion"*

University of Michigan — Ann Arbor, U.S.A.
BSE in Computer Engineering, GPA: 3.9/4.0, Minor in Mathematics. — Sep. 2021 - Apr. 2023

Shanghai Jiaotong University — Shanghai, China
BSE in Electrical and Computer Engineering, GPA: 3.7/4.0, Minor in IP Law — Sep. 2019 - Aug. 2023

Publication

"Overtone Piezoelectric Resonators for Power Conversion" — Aug. 2026
W. Xu, S. Naval, M. Touhami and J. D. Boles, IEEE Transactions on Power Electronics 2026.

"Differential Power Processing Using a Multi-Port Piezoelectric Component" — San Antonio, U.S.A. Mar. 2026
W. Xu, S. Naval and J. D. Boles, Proc. IEEE Applied Power Electronics Conference and Exposition (APEC) 2026.

"An Overtone-Reconfigurable Piezoelectric-Based Power Converter for Wide Load Range Applications" — San Antonio, U.S.A. Mar. 2026
B. Halfmann, **W. Xu** and J. D. Boles, Proc. IEEE Applied Power Electronics Conference and Exposition (APEC) 2026.

"Modeling and Analysis of Piezoelectric-Resonator-Based Power Converters with Finite Terminal Capacitance" — San Antonio, U.S.A. Mar. 2026
B. Karish, **W. Xu** and J. D. Boles, Proc. IEEE Applied Power Electronics Conference and Exposition (APEC) 2026.

- APEC 2026 Technical Session Best Presentation Award by B. Karish*

"High-Efficiency Isolated Piezoelectric Transformers for Magnetic-less DC-DC Power Conversion" — Jan. 2026
S. Naval, **W. Xu**, M. Touhami and J. D. Boles, IEEE Transactions on Power Electronics 2026.

"Approachable Modeling of Multi-port Piezoelectric Components for Power Conversion" — Knoxville, U.S.A. Jun. 2025
W. Xu, S. Naval and J. D. Boles, Proc. IEEE Workshop on Control and Modeling for Power Electronics (COMPEL) 2025.

"Overtone Piezoelectric Transformers for Magnetic-less Power Conversion" — Knoxville, U.S.A. Jun. 2025
S. Naval, **W. Xu**, M. Touhami and J. D. Boles, Proc. IEEE Workshop on Control and Modeling for Power Electronics (COMPEL) 2025.

"High-Efficiency Isolated Piezoelectric Transformers for Magnetic-less DC-DC Power Conversion" — Atlanta, U.S.A.
S. Naval, **W. Xu**, M. Touhami and J. D. Boles, Proc. IEEE Applied Power Electronics Conference and Exposition (APEC) 2025. — Mar. 2025

- APEC 2025 Technical Session Best Presentation Award by S. Naval*

"Overtone Piezoelectric Resonators for Power Conversion" — Lahore, Pakistan
W. Xu, S. Naval, M. Touhami and J. D. Boles, Proc. IEEE Workshop on Control and Modeling for Power Electronics (COMPEL) 2024. — Jun. 2024

- COMPEL 2024 Best Paper Award*

"High-Speed Digital Control in a Switching-Synchronized Sampled-State Space for Variable Frequency Multi-Phase Buck Converters" — Tel-Aviv, Israel
X. Cui, V. Contreras, **W. Xu** and A.-T. Avestruz, Proc. IEEE Workshop on Control and Modeling for Power Electronics (COMPEL) 2022. — Jun. 2022

"Lite-Sparse Hierarchical Partial Power Processing for Heterogeneous Degradation of Batteries in Energy Storage Systems" — Detroit, U.S.A.
A. Ramyar, **W. Xu**, X. Cui, J. Siegel, A. Stefanopoulou and A.-T. Avestruz, Proc. IEEE Energy Conversion Congress and Exposition (ECCE) 2022. — Oct. 2022

Patents

UC Berkeley
U.S. Provisional No. 64/027305 Dickson Switched-Resonator DC-DC Converter — filed 2026

- Inventors:Jessica D. Boles, Wentao Xu

UC Berkeley
International Patent PCT/US2025/033838 Overtone Piezoelectric Resonators for Power Conversion — filed 2026

- Inventors:Jessica D. Boles, Wentao Xu

University of Michigan
International Patent No.PCT/US2023/075703 Power Converter Interconnection — filed 2025

- Inventors: Al-Thaddeus Avestruz, Xiaofan Cui, Veronica Contreras, Wentao Xu

University of Michigan
US Non-provisional Patent Pub.No.20230412061 Power Converter Frequency Control — filed 2024

- Inventors: Al-Thaddeus Avestruz, Xiaofan Cui, Alireza Ramyar, Wentao Xu, Jason Siegel, Anna Stefanopoulou

Scholarships and Honors

UC Berkeley
EECS Evergreen Graduate Research Award (for undergraduate mentorship) — June 2026

UC Berkeley
EECS Demetri Angelakos Memorial Achievement Award — May 2026

Lab-in-fab Piezo MEMS design challenge, announced in IEEE MEMS 2026
Innovation Award — Jan. 2026

IEEE Workshop on Control and Modeling for Power Electronics (COMPEL) 2024
COMPEL 2024 Best Paper Award — Jun. 2024

UC Berkeley
Robert G. Meyer Fellowship — Sep. 2023

University of Michigan
EECS Undergraduate Outstanding Research Award — Apr. 2023

Shanghai Jiaotong University
Shanghai Jiaotong University Junyuan Scholarships — Oct. 2022, 2021, 2020

University of Michigan
Roger King Scholarship — Aug. 2022

University of Michigan
Cheng Family Scholarship — Aug. 2021

2020 University Physics Competition
Silver Medal — Jan. 2021

Shanghai Jiaotong University
John & Jane Sunshine Scholarship — Dec. 2020